ROVER CHALLENGE

WITH AWS JUPYTPER'S NOTEBOOK

Presented By: Gavi, George, & Jocelyn

ABSTRACT

Our focus: Build an algorithm to quickly navigate a rover along a complex path to the location of an injured firefighter, who needs assistance. The data package includes high-resolution NAIP imagery, and an accurate road network for the same area.

Methodology and Results Found

Step 1:

- Plot out Guidance, Navigation, and Control System Algorithm as a pseduo-code for the robot's trajectory and method of movement

Step 2:

- Converted relevant tif files to manipulate arrays to represent obstacles and traversable data points for a path

Step 3:

- Use rasterio package functions to derive arrays of correct dimensions from imagery data
  - 0 = traversable
  - 1 = obstacle
  - 2 = path

Step 4:

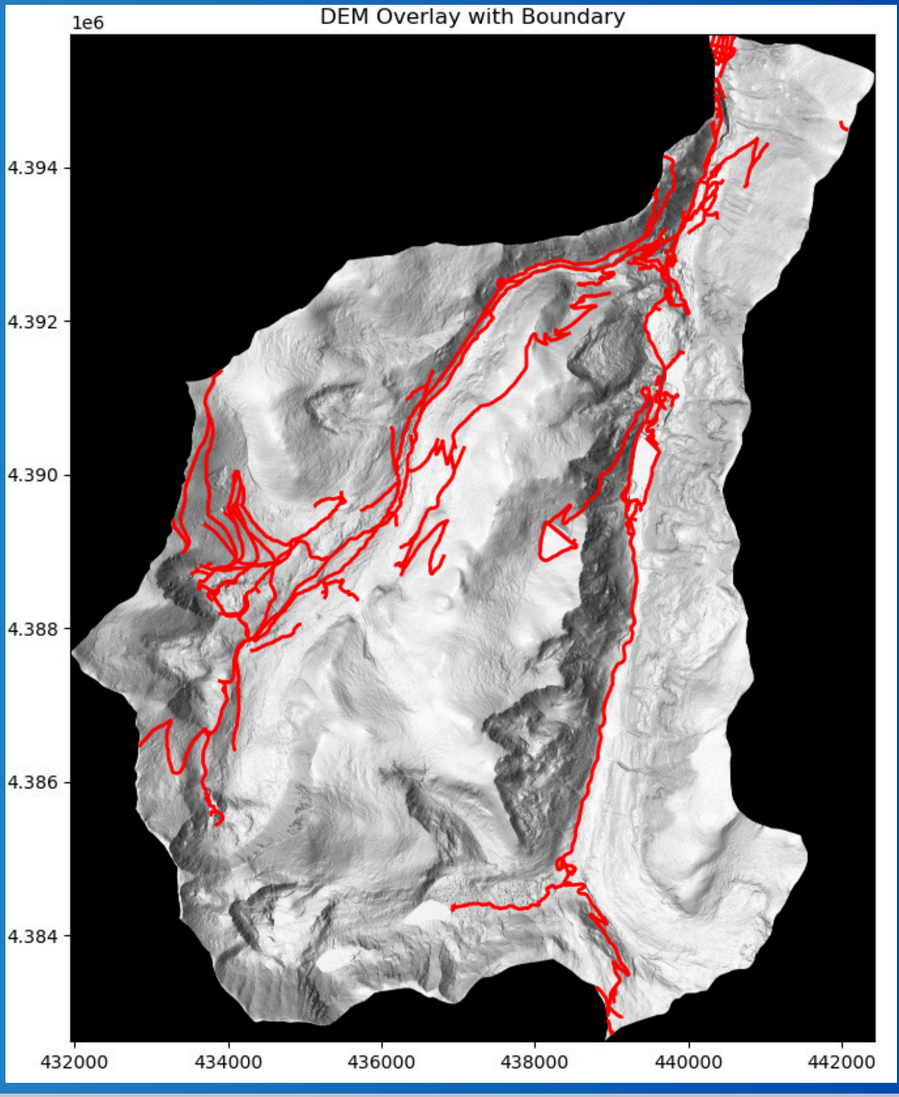
- Extract each band within the data into 2-D array

Step 5:

- Generated random coordinates within graph to become obstacles
  - Rand function and loop to assign indices to 0

Step 6:

- Path coordinate values overrides any obstacles
  - Create a new path for moving around obstacle



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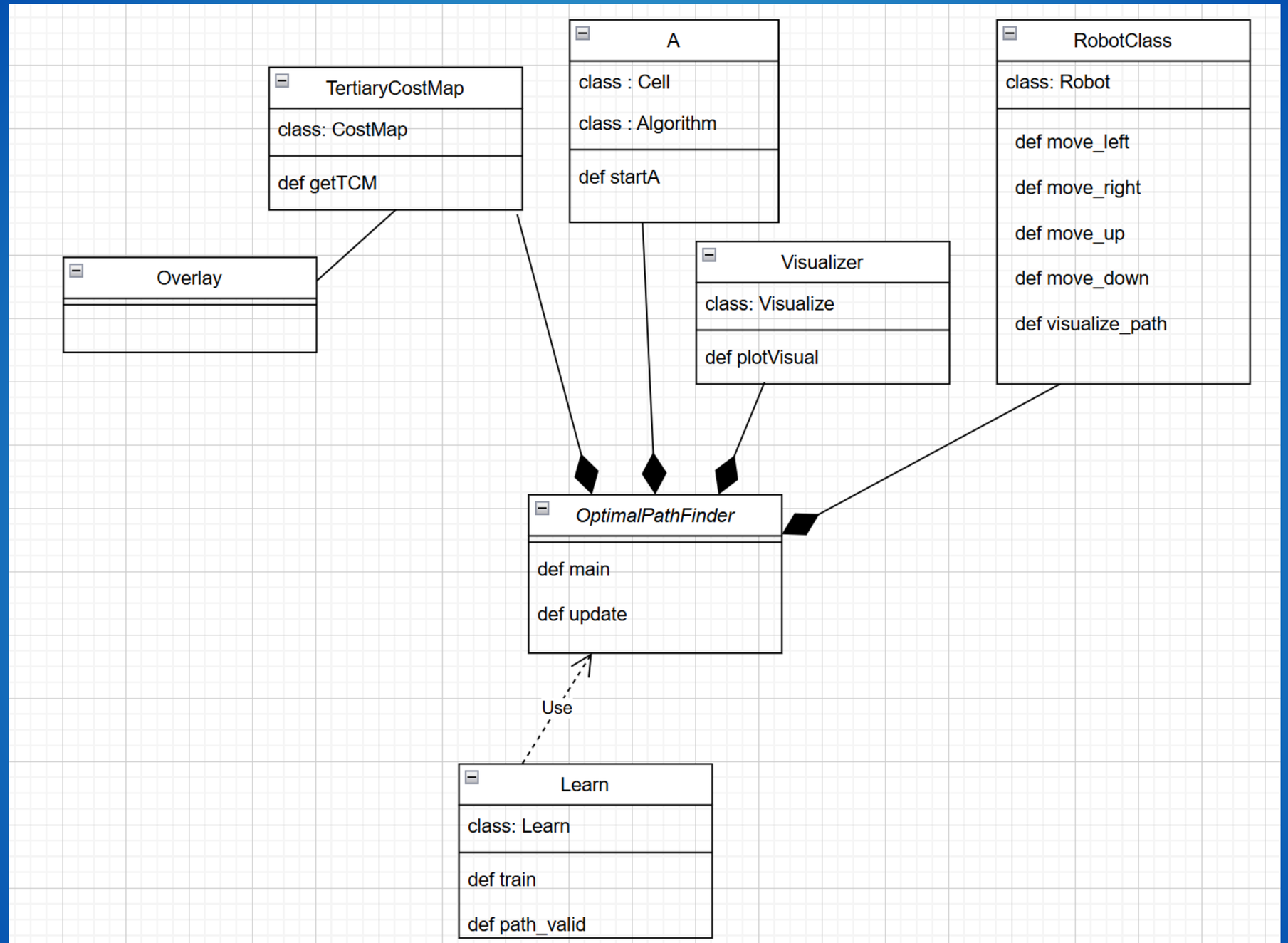
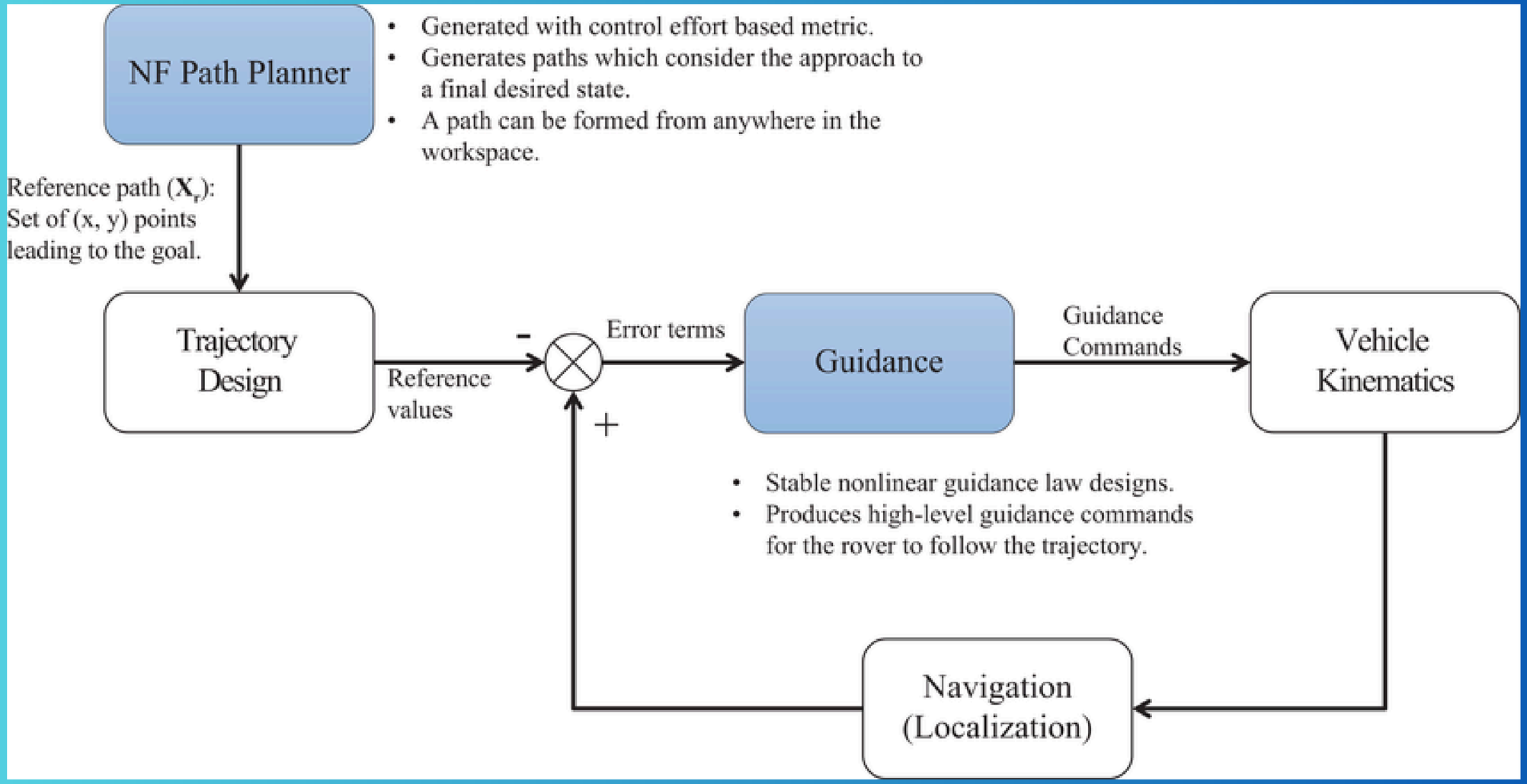

A
class : Cell
class : Algorithm
def startA
RobotClass
class: Robot
def move_left
def move_right
def move_up
def move_down
def visualize_path
TertiaryCostMap
class: CostMap
def getTCM
Overlay
Visualizer
class: Visualize
def plotVisual
OptimalPathFinder
def main
def update
Use
Learn
class: Learn
def train
def path_valid

Design Choices and Rationale



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Team Collaboration and Role Distribution

- **Gavi:**
  - Creating images with distinct paths
  - Robot algorithm
- **George:**
  - Refactoring from pygame to matplotlib
  - Managing the environment
- **Jocelyn:**
  - Algorithm and logic/decision-making
  - Design/Structure

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Packages and Libraries Used

- **Matplotlib**
  - **Replacement for pygame**
- **Rasterio**
  - **Handling images and data**
- **Numpy**
  - **Creating arrays and calculation**
- **GeoPandas**
  - **Geological data handling**
- **Other: math and heapq**

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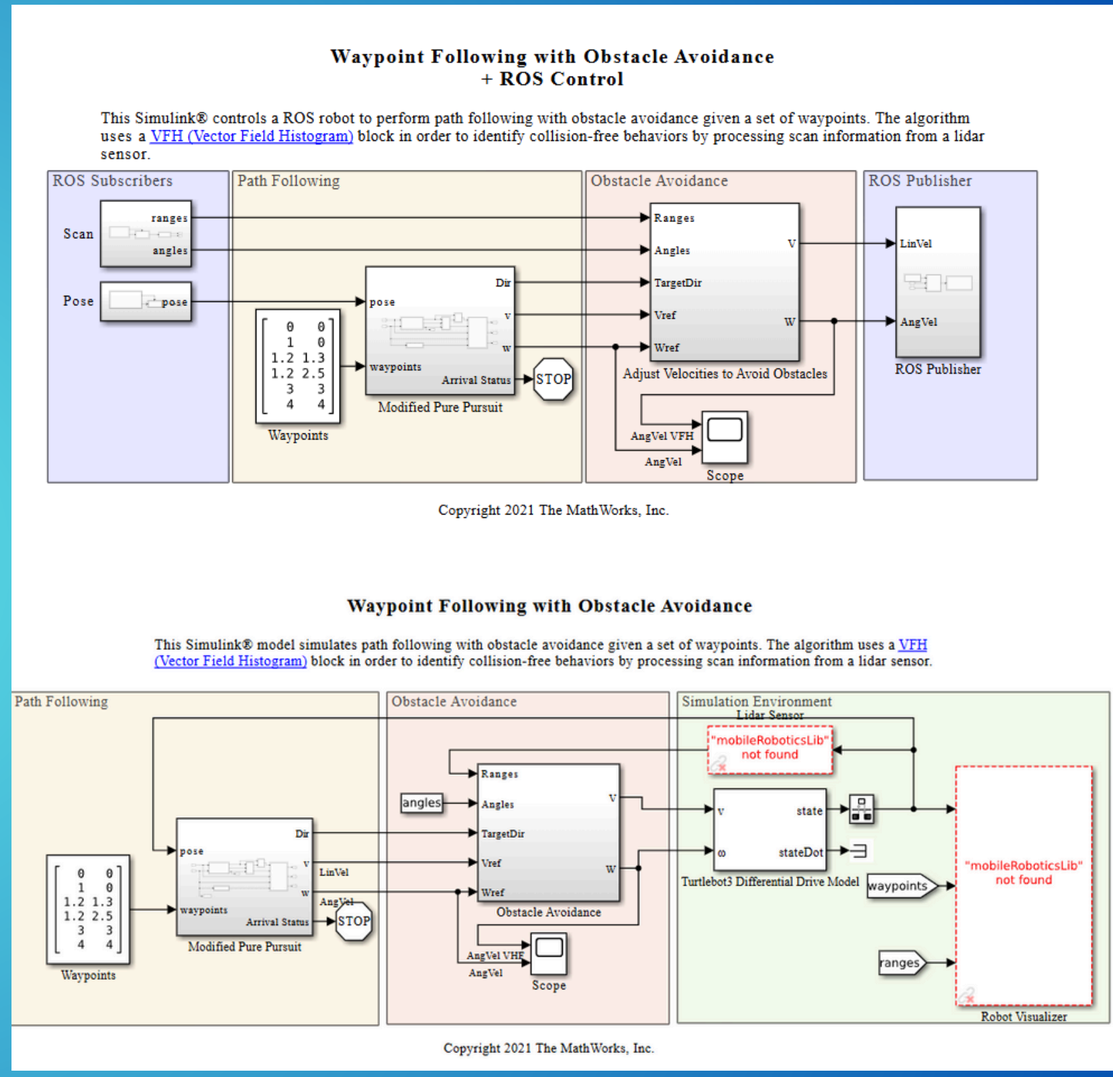
Innovative Approaches

Robot's algorithm:

- **Researched different navigation and guidance models for mobile robots and unmanned ground vehicles (UGV) in Matlab and other platforms (such as A*)**
- **Researched ROS (Robot Operating System) enabled simulations for autonomous navigation of Turtlebot3 with obstacle avoidance in Matlab**
  - **createImageMask**
  - **convertMapToMatrix**
  - **LidarSlam**
  - **LidarSlam_interp**
- **This improves efficiency, reliability, and the ability to automate complex tasks for real-life implementation**



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Innovative Approaches

Design choice utilizing matrices

Dedicating values within an array to hold significance, rather than creating multiple data structures.

This:

- **Saves time**
- **Saves allocation on the heap and stack**

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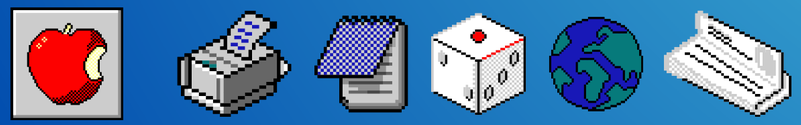
Factors Considered in the Model

- If there is an obstacle on the direct path, is there traversable path around we can use to avoid the object?
- What are the most accurate and time-efficient pathways that the robot can travel?
- What timesteps are needed for the robot to collect travel data?
- How to determine steep elevation changes and consider them as untraversable?
- How do those road boundary and DEM files provided work together to identify coordinates



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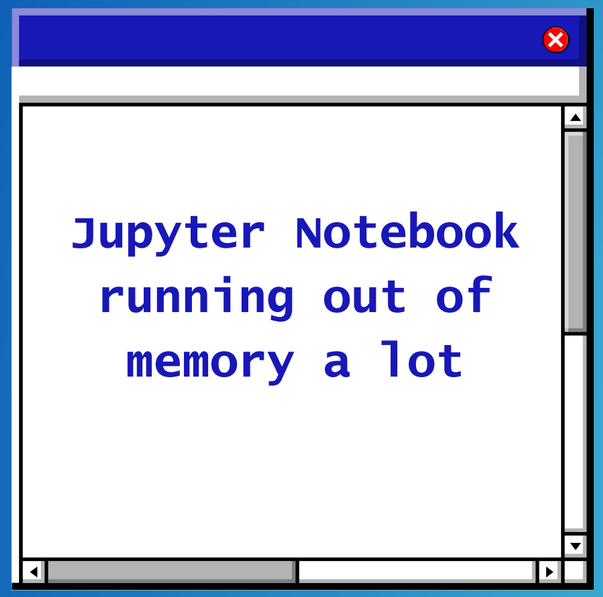
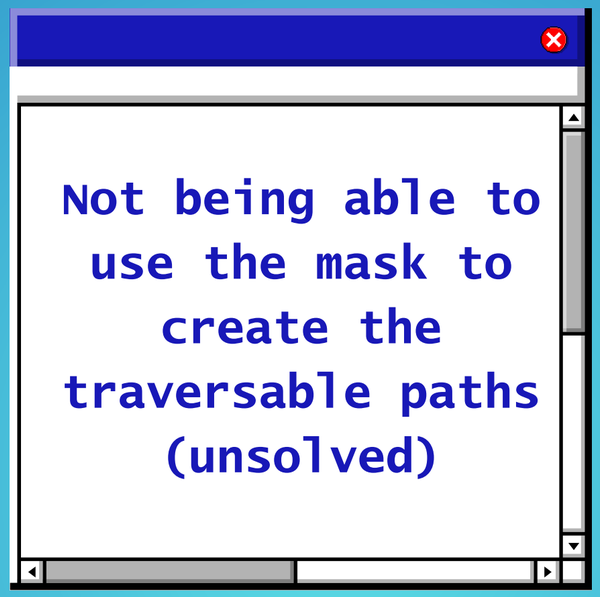
Challenges Faced and Lessons Learned





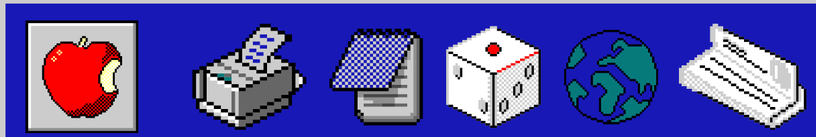
Lessons Learned

- **Some solutions may come with drawbacks**
- **More applications to Linear Algebra**

What We Would Do Differently

- Change to the GPU profile earlier
  - ***Calculating and maintaining a prediction of time and problem complexity in order to anticipate these changes***

- AND Deciding to change direction early on if something isn’t working (be less stubborn!)



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Future Development Plans with Additional Time

Use documented robotic navigation schemes such as:

- **Implement Monte Carlo Localization (MCL) with Optimized Path**
  - **Determines moving vehicle's position within an environment using a map, sensors, and a set of random samples (particles) to represent the robot's belief about state.**
  - **Helps in interpreting noisy sensor data, by generating multiple hypotheses about the world state and updating beliefs based on the likelihood of sensor readings.**
- **Focus more on iterating the model**
  - **Machine Learning training optimization**

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Thank you to NASA, USDA, US Forest Service, AWS, and CSU for this amazing opportunity!

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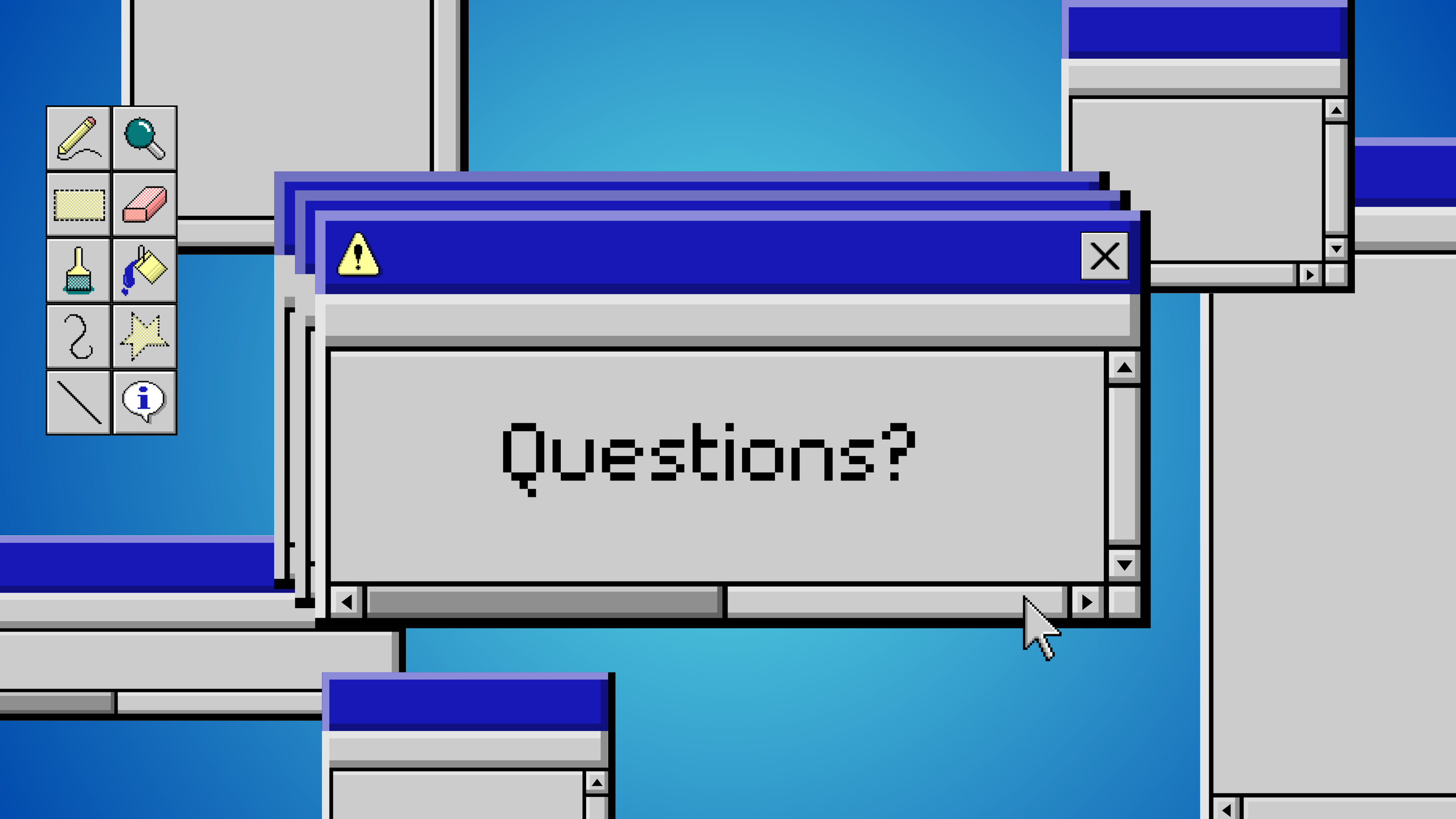
Questions?

Works Cited

- Matlab’s Monte Carlo Localization Algorithm
- Image for GNC Flowchart
- Simulinks for Autonomous Navigation for Mobile Robots and UGV
- A Algorithm

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